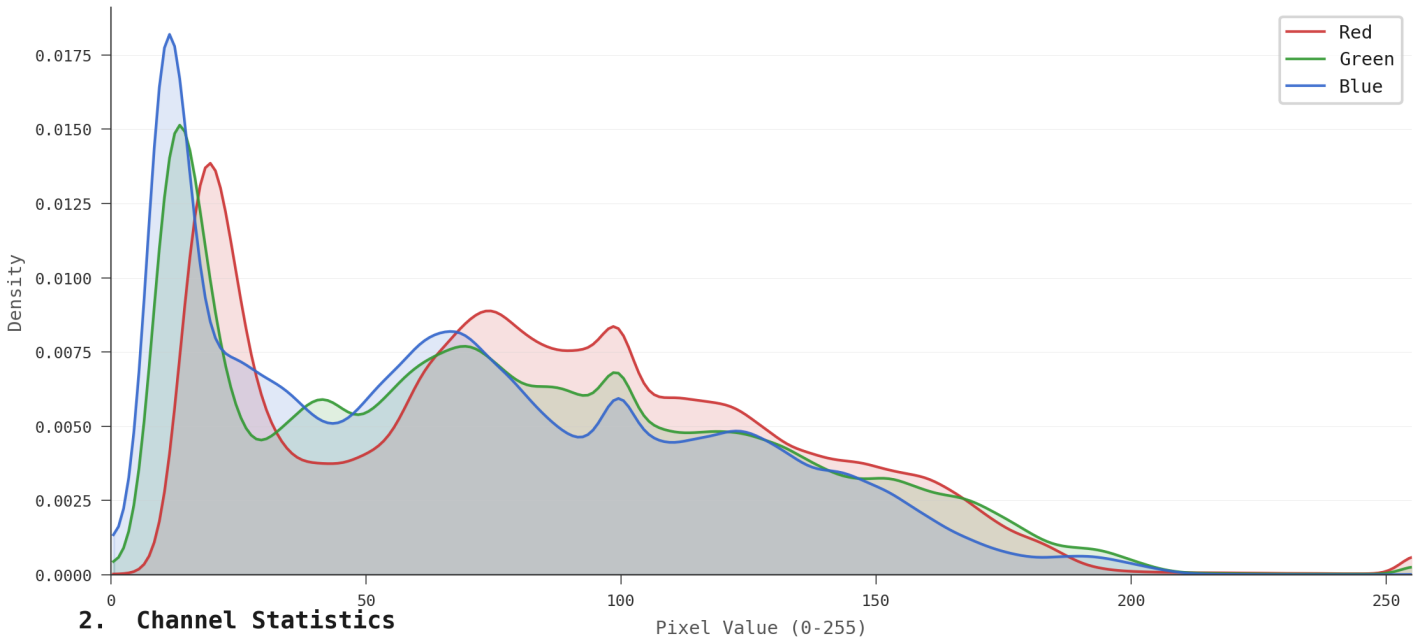
Reference Data Sheet: KODIM17

Subject: Living Statue Street Performer

512x768 | 393,216 px
24-bit RGB | sRGB | PNG (lossless)

1. RGB Channel Distribution



2. Channel Statistics

Channel	Mean	Std Dev	Variance	Kurtosis	Skewness	Min	Max
Red	82.6843	46.9409	2203.4435	-0.4375	0.3475	1	255
Green	77.0529	50.1148	2511.4934	-0.6389	0.4340	0	255
Blue	69.0200	47.9270	2296.9984	-0.6750	0.4952	0	255

3. Inter-Channel Correlation

	R	G	B
R	1.0000	0.9852	0.9334
G	0.9852	1.0000	0.9777
B	0.9334	0.9777	1.0000

Pairwise Pearson Correlation Coefficients

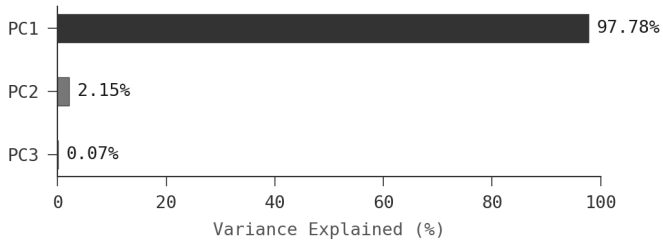
r(R-G)	=	**0.985194**
r(R-B)	=	**0.933405**
r(G-B)	=	**0.977749**
Avg |r|	=	**0.965449**

4. Covariance Matrix S

$$
\begin{array}{c} \\ R \\ G \\ B \end{array}
\begin{array}{c}
\begin{array}{ccc} R & G & B \end{array} \\
\begin{bmatrix}
\mathbf{2203.4491} & 2317.6074 & 2099.9185 \\
2317.6074 & \mathbf{2511.4998} & 2348.4144 \\
2099.9185 & 2348.4144 & \mathbf{2297.0042}
\end{bmatrix}
\end{array}
$$

Reference Data Sheet: KODIM17

5. Eigendecomposition



Eigenvector Loadings

	R	G	B	Eigenvalue
PC1	-0.5578	**-0.6048**	-0.5684	6856.2614
PC2	-0.6771	-0.0645	0.7330	150.7549
PC3	0.4800	-0.7938	0.3735	4.9369

6. Derived Metrics

Condition Number

1388.78

(l1/l3) PC1/PC2: 45.48 | PC2/PC3: 30.54

Blue Channel Independence

3.6%

Fraction of blue variance not captured by PC1

PC1 Dominant Channel

Green

Loading: -0.6048 | Green dominant

Classification: Near-Degenerate (PC1 = 97.78%)

7. Spatial Autocorrelation (lag-1)

Channel	Horizontal	Vertical	Average
Red	0.972133	0.969784	0.970959
Green	0.972996	0.970793	0.971894
Blue	0.971203	0.969172	0.970187

8. Average Local Variance (3x3)

Channel	Value
Red	99.4767
Green	111.5929
Blue	105.5583

9. Redundancy Profile

Classification: Near-Degenerate | Condition Number: 1388.78 | PC1: 97.78%

Correlation: R–G pair dominates at r = 0.985194 | G–B: 0.977749 | R–B: 0.933405

Blue Channel: 3.6% independent | 96.4% captured by PC1

Spatial: Avg autocorrelation: 0.971013 | Avg local variance: 105.5426

Eigenvector: Green dominant | G (-0.6048) leads | R (-0.5578), B (-0.5684)

References

[1] Eastman Kodak Company. Kodak Publication No. PCD-042, 1992.

[2] Baetzel, J. (2026). "Statistical Characterization of Inter-Channel Redundancy Structure in the Kodak Lossless True Color Image Suite."

[3] Watanabe, S. "Karhunen-Loeve Expansion and Factor Analysis," pp. 635-660, 1965.

[4] Giorgianni, E.J. and Madden, T.E. Digital Color Management. Addison-Wesley, 1998.

Blue Channel Independence = (1 - (blue_loading_PC1^2 x l1 / Var(B))) x 100

Data Source: Statistics derived from Baetzel (2026), "Statistical Characterization of Inter-Channel Redundancy Structure in the Kodak Lossless True Color Image Suite."